Survival Analysis

From the Same `.qmd` Source — Now as Slides

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Table of contents I

Section 1

Background

Motivation

- Survival data arise in clinical trials, epidemiology, economics
- Outcome: time until an event (death, relapse, failure)
- Key challenge: **censoring** — some subjects do not experience the event during follow-up

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Goal: estimate the relationship between covariates and survival time.

The Cox Model

The Cox proportional hazards model (Cox 1972):

$$h(t \mid \mathbf{x}) = h_0(t)\exp(\mathbf{x}^\top \beta)$$

Advantages

- Semi-parametric: $h_0(t)$ unspecified
- Interpretable hazard ratios
- Handles censoring naturally

Assumptions

- Proportional hazards
- Log-linearity
- Independence

Section 2

Data and Methods

Dataset

```r
data(lung, package = "survival")
dplyr::glimpse(lung)
```

```
Rows: 228
Columns: 10
$ inst      <dbl> 3, 3, 3, 5, 1, 12, 7, 11, 1, 7, 6, 16, 11, 2
$ time      <dbl> 306, 455, 1010, 210, 883, 1022, 310, 361, 21
$ status    <dbl> 2, 2, 1, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2,
$ age       <dbl> 74, 68, 56, 57, 60, 74, 68, 71, 53, 61, 57,
$ sex       <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 2, 2, 1, 1,
$ ph.ecog   <dbl> 1, 0, 0, 1, 0, 1, 2, 2, 1, 2, 1, 2, 1, NA, 1
$ ph.karno  <dbl> 90, 90, 90, 90, 100, 50, 70, 60, 70, 70, 80,
$ pat.karno <dbl> 100, 90, 90, 60, 90, 80, 60, 80, 80, 70, 80,
$ meal.cal  <dbl> 1175, 1225, NA, 1150, NA, 513, 384, 538, 825
$ wt.loss   <dbl> NA, 15, 15, 11, 0, 0, 10, 1, 16, 34, 27, 23,
```

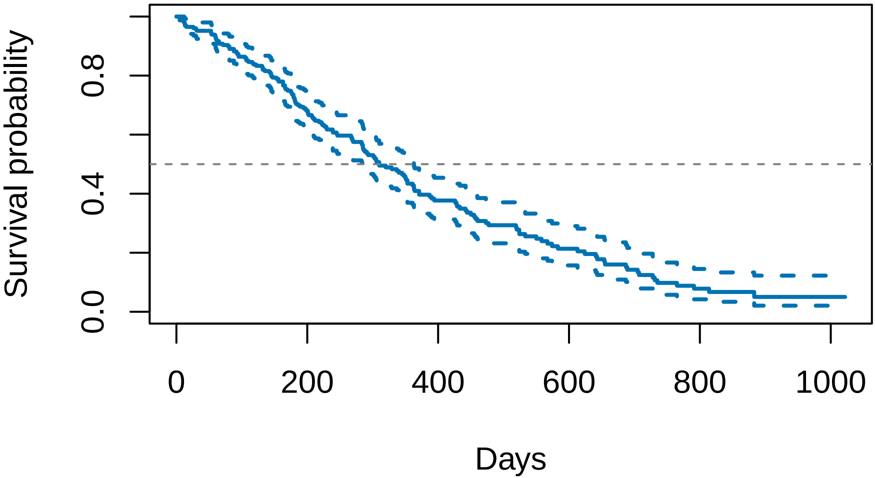

Figure 1: Overall Kaplan-Meier survival estimate.

Survival by Sex

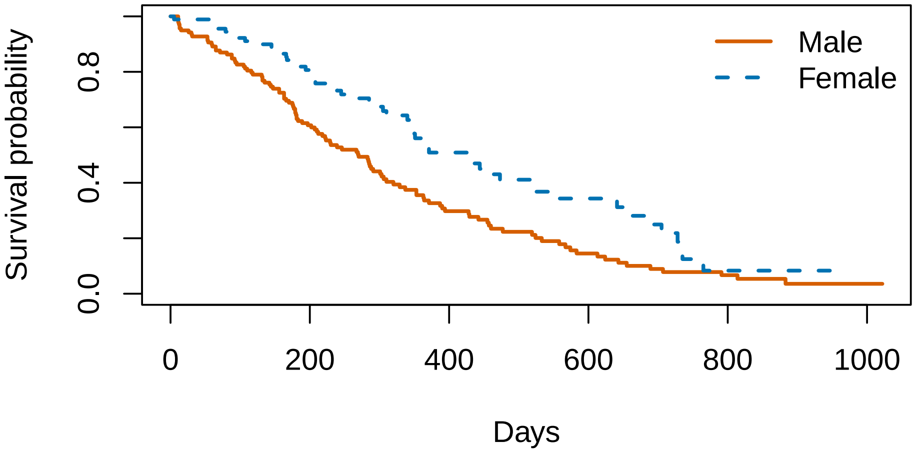

Figure 2: Kaplan-Meier curves by sex.

Section 3

Results

Cox Regression Results

Table 1: Cox model hazard ratios.

Variable	HR	CI Low	CI High	p
age	1.011	0.993	1.030	0.232
sex	0.575	0.414	0.799	0.001
ph.ecog	1.590	1.273	1.986	0.000

Key Findings

- Older age associated with **worse** survival (HR $= 1.01$ per year)
- Female sex associated with **better** survival (HR $= 0.59$)
- Higher ECOG score associated with **worse** survival

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Note

These results replicate the classic findings from the NCCTG study.

Section 4

Conclusion

Take-Home Messages

1. Quarto renders the **same `.qmd` file** to PDF, HTML, and slides
2. Code + prose stay together — no copy-paste
3. All numbers, figures, and tables update on re-render
4. Switch output format by changing one line in the YAML

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```
quarto render slides.qmd --to beamer   # PDF slides
quarto render slides.qmd --to revealjs # HTML slides
quarto render slides.qmd --to pdf      # article PDF
```

References

Cox, D. R. 1972. "Regression Models and Life-Tables." *Journal of the Royal Statistical Society: Series B* 34 (2): 187–220. https://doi.org/10.1111/j.2517-6161.1972.tb00899.x.